

MATH 3032 — Mathematical Statistics

Chapter 3: Point Estimation

Complete Exam Preparation Guide

All 8 Exam Questions · Worked Examples · Practice Problems · Mock Exam

Topics covered

- Q1 Likelihood function $L(\theta)$
- Q2 MLE $\hat{\theta}$ + second derivative test
- Q3 Bias via $\mathbb{E}[X_1]$ and u-substitution
- Q4 Mean Squared Error
- Q5 Consistency (LLN and Chebyshev)
- Q6 Fisher information $I(\theta)$
- Q7 Cramér-Rao lower bound
- Q8 Asymptotic normality

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Textbooks:

Panaretos, *Statistics for Mathematicians* (Ch. 3)

Hogg, Tanis & Zimmerman, *Probability and Statistical Inference*, 9th ed.
(\$6.4, \$6.6)

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Textbooks: Panaretos, *Statistics for Mathematicians* (Ch. 3) & Hogg–Tanis–Zimmerman, *Probability and Statistical Inference*, 9th ed. (§6.4, §6.6)

Exam structure: 8 questions on a single continuous pdf, 30 points total. Questions 1–2 are $\approx \frac{1}{3}$ of points; Questions 3–5 are $\approx \frac{1}{3}$; Questions 6–8 are $\approx \frac{1}{3}$.

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The Statistical Framework

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_\theta$ with pdf $f(x; \theta)$, $\theta \in \Theta \subset \mathbb{R}$ unknown. The goal of Chapter 3 is to:

1. Construct an estimator $\hat{\theta}$ — a function of the sample that approximates θ .
2. Analyse its quality: bias, mean squared error, consistency.
3. Assess optimality: does $\hat{\theta}$ achieve the Cramér-Rao lower bound?
4. Describe its large-sample distribution.

$$\begin{aligned} \int_0^\infty e^{-u} du &= 1, & \int_0^\infty u e^{-u} du &= 1, & \int_0^\infty u^2 e^{-u} du &= 2 \\ \int_0^\infty u^3 e^{-u} du &= 6, & \int_0^\infty u^4 e^{-u} du &= 24, & \int_0^\infty u^n e^{-u} du &= n! \end{aligned}$$

Also: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$; $\text{Var}(aX) = a^2 \text{Var}(X)$; $\text{Var}(\sum X_i) = n \text{Var}(X_1)$ if i.i.d.

The Eight Exam Questions — Theory

Q2.1. Likelihood Function $L(\theta)$

Given a pdf $f(x; \theta)$, write the likelihood function $L(\theta)$ for an i.i.d. sample $X_1, \dots, X_n$ and simplify.

Definition. The likelihood function is the joint pdf of the sample, viewed as a function of θ :

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta).$$

Since the samples are i.i.d., the joint pdf factors into a product of marginals. The key simplification steps are:

1. Write out all n factors explicitly.
2. Group **constants** (terms involving only θ): $\prod c(\theta) = c(\theta)^n$.
3. Group **data terms** (products of functions of X_i): $\prod g(X_i)$.
4. **Collapse exponentials**: $\prod e^{h(X_i, \theta)} = \exp(\sum h(X_i, \theta))$.

Always identify the *exponent's structure* in $f(x; \theta)$ first. If the exponent is $-x/\theta$, then L will contain $e^{-\sum X_i/\theta}$. If the exponent is $-x^2/\theta^2$, then L will contain $e^{-\sum X_i^2/\theta^2}$. This exponent determines which sufficient statistic appears in $\hat{\theta}$.

Review-session worked example. Let $f(x; \theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$, $x > 0$. Then:

$$L(\theta) = \prod_{i=1}^n \frac{X_i^2 e^{-X_i/\theta}}{2\theta^3} = \underbrace{\frac{1}{(2\theta^3)^n}}_{\text{constant}^n} \cdot \underbrace{\prod_{i=1}^n X_i^2}_{\text{data}} \cdot \underbrace{\exp\left(-\frac{\sum X_i}{\theta}\right)}_{\text{collapsed exp.}}$$

Q2.2. Maximum Likelihood Estimator $\hat{\theta}$

Find $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta)$. **Justify that it is a maximum** using the second derivative test.

Method.

1. Take the **log-likelihood** $\ell(\theta) = \log L(\theta)$. Since log is strictly increasing, $\arg \max L = \arg \max \ell$.
2. **Differentiate** $\ell(\theta)$ w.r.t. θ and set $\ell'(\theta) = 0$ (score equation).
3. **Solve** for $\hat{\theta}$.
4. **Second derivative test**: evaluate $\ell''(\hat{\theta})$. If $\ell''(\hat{\theta}) < 0$, then $\hat{\theta}$ is a local maximum.

After finding $\hat{\theta}$, the score equation $\ell'(\hat{\theta}) = 0$ gives a relation of the form $\sum(\text{data}) = n \cdot g(\hat{\theta})$ for some function g . Substitute this into $\ell''(\hat{\theta})$ to simplify the expression and determine its sign without knowing the data values explicitly.

Review-session worked example. Continuing from Q1:

$$\begin{aligned} \ell(\theta) &= -n \log(2\theta^3) + 2 \sum \log X_i - \frac{\sum X_i}{\theta} \\ \ell'(\theta) &= -\frac{3n}{\theta} + \frac{\sum X_i}{\theta^2} = 0 \implies \hat{\theta} = \frac{\sum X_i}{3n} = \frac{\bar{X}}{3}. \end{aligned}$$

Second derivative test:

$$\ell''(\theta) = \frac{3n}{\theta^2} - \frac{2 \sum X_i}{\theta^3}.$$

From the score equation: $\sum X_i = 3n\hat{\theta}$. Substituting:

$$\ell''(\hat{\theta}) = \frac{3n}{\hat{\theta}^2} - \frac{2 \cdot 3n\hat{\theta}}{\hat{\theta}^3} = \frac{3n}{\hat{\theta}^2} - \frac{6n}{\hat{\theta}^2} = -\frac{3n}{\hat{\theta}^2} < 0. \quad \checkmark$$

Q2.3. Bias of $\hat{\theta}$

Compute $\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta}] - \theta$. This requires evaluating $\mathbb{E}[X_1]$ by integration.

Method.

1. Use **linearity** to reduce $\mathbb{E}[\hat{\theta}]$ to $\mathbb{E}[X_1]$: e.g. if $\hat{\theta} = \frac{1}{cn} \sum X_i$, then $\mathbb{E}[\hat{\theta}] = \frac{\mathbb{E}[X_1]}{c}$.
2. Compute $\mathbb{E}[X_1] = \int_0^\infty x f(x; \theta) dx$ via **u-substitution** $u = x/\theta$ (or $u = x^2/\theta^2$ for Weibull-type pdfs).

3. Apply the gamma integral formula from the formula sheet.
4. Compute Bias = $\mathbb{E}[\hat{\theta}] - \theta$.

Review-session worked example. With $f(x; \theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$ and $\hat{\theta} = \bar{X}/3$:

$$\mathbb{E}[X_1] = \int_0^\infty x \cdot \frac{x^2 e^{-x/\theta}}{2\theta^3} dx = \frac{1}{2\theta^3} \int_0^\infty x^3 e^{-x/\theta} dx.$$

Substitute $u = x/\theta$, so $x = \theta u$, $dx = \theta du$:

$$= \frac{1}{2\theta^3} \cdot \theta^4 \int_0^\infty u^3 e^{-u} du = \frac{\theta}{2} \cdot 3! = \frac{\theta}{2} \cdot 6 = 3\theta.$$

Therefore $\mathbb{E}[\hat{\theta}] = \frac{3\theta}{3} = \theta$, so Bias($\hat{\theta}, \theta$) = 0. $\hat{\theta}$ is **unbiased**.

For Gamma-type pdfs with pdf $\propto x^{\alpha-1} e^{-x/\theta}$: the integral $\mathbb{E}[X_1]$ uses the ($n = \alpha$) row of the table, and $\mathbb{E}[X_1^2]$ uses the ($n = \alpha + 1$) row. In general, $\mathbb{E}[X_1^k]$ uses row $n = \alpha + k - 1$, giving value $(\alpha + k - 1)!$. For Weibull-type pdfs with e^{-x^2/θ^2} in the exponent, use $u = x^2/\theta^2$ instead.

Q2.4. Mean Squared Error

Compute $\text{MSE}(\hat{\theta}, \theta) = \mathbb{E}[(\hat{\theta} - \theta)^2]$.

Decomposition (Panaretos Remark 3.5):

$$\text{MSE}(\hat{\theta}, \theta) = \text{Bias}(\hat{\theta}, \theta)^2 + \text{Var}(\hat{\theta}).$$

If Bias = 0, then $\text{MSE} = \text{Var}(\hat{\theta})$.

Method for computing $\text{Var}(\hat{\theta})$:

1. Pull the constant out squared: $\text{Var}(\frac{1}{cn} \sum X_i) = \frac{1}{c^2 n^2} \cdot n \text{Var}(X_1) = \frac{\text{Var}(X_1)}{c^2 n}$.
2. Compute $\text{Var}(X_1) = \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2$.
3. $\mathbb{E}[X_1^2]$ requires the same u-substitution, using the next row of the gamma table.

The professor's explicit rule: your MSE *must* have n in the denominator. If it doesn't, you dropped the n somewhere in the variance-of-sum step: $\text{Var}(\sum X_i) = n \text{Var}(X_1)$, so $\text{Var}(\hat{\theta}) = \text{Var}(X_1)/(c^2 n)$, not $\text{Var}(X_1)/c^2$.

Review-session worked example. With $\mathbb{E}[X_1] = 3\theta$:

$$\begin{aligned} \mathbb{E}[X_1^2] &= \frac{1}{2\theta^3} \int_0^\infty x^4 e^{-x/\theta} dx = \frac{\theta^2}{2} \int_0^\infty u^4 e^{-u} du = \frac{\theta^2}{2} \cdot 4! = \frac{\theta^2}{2} \cdot 24 = 12\theta^2. \\ \text{Var}(X_1) &= 12\theta^2 - (3\theta)^2 = 12\theta^2 - 9\theta^2 = 3\theta^2. \\ \text{Var}(\hat{\theta}) &= \frac{1}{(3n)^2} \cdot n \cdot 3\theta^2 = \frac{3n\theta^2}{9n^2} = \frac{\theta^2}{3n}. \\ \text{MSE}(\hat{\theta}, \theta) &= 0 + \frac{\theta^2}{3n} = \boxed{\frac{\theta^2}{3n}}. \end{aligned}$$

Q2.5. Consistency

Show that $\hat{\theta} \xrightarrow{p} \theta$ as $n \rightarrow \infty$.

There are two approaches; choose based on $\hat{\theta}$'s structure.

Approach 1 — Law of Large Numbers. If $\hat{\theta} = \frac{1}{n} \sum g(X_i)$:

$$\text{By LLN: } \frac{1}{n} \sum g(X_i) \xrightarrow{p} \mathbb{E}[g(X_1)] = \theta. \quad \checkmark$$

Requires: $\mathbb{E}[|g(X_1)|] < \infty$ (verified since we computed $\mathbb{E}[X_1] < \infty$).

Approach 2 — Chebyshev inequality. Always valid when MSE is computed:

$$P(|\hat{\theta} - \theta| \geq \varepsilon) \leq \frac{\text{MSE}(\hat{\theta}, \theta)}{\varepsilon^2} = \frac{\theta^2}{3n\varepsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

By the squeeze theorem, $\hat{\theta} \xrightarrow{p} \theta$. $\checkmark$

LLN: use when $\hat{\theta}$ is visibly $(1/n) \sum g(X_i)$. Cleaner write-up.

Chebyshev: universal fallback — use whenever MSE is already computed (i.e. Q4 is done).

Also handles cases where $\hat{\theta}$ is not a simple average (e.g. $\hat{\theta} = \sqrt{\sum X_i^2/n}$).

Key insight (Panaretos Remark 3.8): $\text{MSE}(\hat{\theta}, \theta) \rightarrow 0 \Rightarrow \hat{\theta} \xrightarrow{p} \theta$.

Q2.6. Fisher Information $I(\theta)$

Compute $I(\theta) = \mathbb{E}_{\theta} \left[\left(\frac{\partial}{\partial \theta} \log f(X_1; \theta) \right)^2 \right]$.

Method.

1. Compute $\log f(x; \theta)$ — this is the $n = 1$ case of the log-likelihood.
2. Differentiate in θ to get the **score** $s(\theta) = \frac{\partial}{\partial \theta} \log f(X_1; \theta)$.
3. Compute $I(\theta) = \mathbb{E}[s(\theta)^2]$.

Under regularity conditions, $\mathbb{E}[s(\theta)] = 0$, so $I(\theta) = \mathbb{E}[s^2] = \text{Var}(s)$. If the score takes the form

$$s(\theta) = \frac{X_1 - \mathbb{E}[X_1]}{c(\theta)},$$

then immediately $I(\theta) = \text{Var}(X_1)/c(\theta)^2$ — no new integrals needed. This pattern appears whenever the log-likelihood is linear in X_1 .

Alternative formula (negative second derivative):

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X_1; \theta) \right].$$

Review-session worked example.

$$\log f = -\log(2\theta^3) + 2 \log X_1 - \frac{X_1}{\theta}, \quad s(\theta) = \frac{X_1}{\theta^2} - \frac{3}{\theta} = \frac{X_1 - 3\theta}{\theta^2}.$$

Since $\mathbb{E}[X_1] = 3\theta$, the score is $s(\theta) = (X_1 - \mathbb{E}[X_1])/\theta^2$. Therefore:

$$I(\theta) = \frac{\text{Var}(X_1)}{\theta^4} = \frac{3\theta^2}{\theta^4} = \frac{3}{\theta^2}.$$

Q2.7. Cramér-Rao Lower Bound

Does $\hat{\theta}_{\text{MLE}}$ attain the Cramér-Rao lower bound?

Theorem 3.9 (Cramér-Rao, Panaretos). Let T be any estimator with bias $\beta(\theta) = \mathbb{E}[T] - \theta$. Under regularity conditions:

$$\text{Var}(T) \geq \frac{(1 + \beta'(\theta))^2}{n \cdot I(\theta)}.$$

For **unbiased** estimators ($\beta = 0, \beta' = 0$):

$$\text{Var}(T) \geq \frac{1}{n \cdot I(\theta)} =: \text{CRLB}.$$

For estimating a function $g(\theta)$ (e.g. $g(\theta) = \theta^2$):

$$\text{Var}(T) \geq \frac{[g'(\theta)]^2}{n \cdot I(\theta)}.$$

Review-session worked example.

$$\text{CRLB} = \frac{1}{n \cdot I(\theta)} = \frac{1}{n \cdot 3/\theta^2} = \frac{\theta^2}{3n}.$$

Since $\text{Var}(\hat{\theta}) = \frac{\theta^2}{3n} = \text{CRLB}$, the MLE **attains the Cramér-Rao bound**. Therefore $\hat{\theta}$ is the UMVUE — no unbiased estimator has smaller variance.

For $f(x; \theta) = 1/\theta, 0 < x < \theta$: the support depends on θ , violating a regularity condition. The score $s(\theta) = -1/\theta$ contains no X_1 , so $\mathbb{E}[s] \neq 0$. The MLE is $\hat{\theta} = X_{(n)} = \max(X_1, \dots, X_n)$, which is biased: $\mathbb{E}[X_{(n)}] = n\theta/(n+1)$. The bias-corrected estimator $T^* = \frac{n+1}{n}X_{(n)}$ has $\text{Var}(T^*) = \theta^2/(n(n+2))$, which is *strictly less than* the formal CRLB θ^2/n — demonstrating that the bound is invalid for this model.

Q2.8. Asymptotic Normality

Find σ such that $\frac{\sqrt{n}}{\sigma}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1)$.

Approach 1 — Classical CLT. If $\hat{\theta} = \frac{1}{n} \sum g(X_i)$:

$$\frac{\sqrt{n}(\hat{\theta} - \theta)}{\sqrt{\text{Var}(g(X_1))}} \xrightarrow{d} N(0, 1), \quad \sigma = \sqrt{\text{Var}(g(X_1))}.$$

Approach 2 — MLE CLT (Panaretos §3.3.3). For any regular MLE:

$$\sqrt{n \cdot I(\theta)}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1), \quad \sigma = \frac{1}{\sqrt{I(\theta)}}.$$

Both give $\sigma^2 = 1/I(\theta)$, and both equal $\text{Var}(g(X_1))$, only when:

$$\text{Var}(g(X_1)) = \frac{1}{I(\theta)} \iff \text{Var}(\hat{\theta}) = \frac{1}{n \cdot I(\theta)} = \text{CRLB}.$$

This three-way identity $\text{Var}(g(X_1)) = 1/I(\theta) = n \cdot \text{Var}(\hat{\theta})$ holds *if and only if* the MLE attains the Cramér-Rao bound. When the bound is not attained, the two approaches give different values of σ , and the MLE CLT (Approach 2) may not apply in its standard form.

Review-session worked example.

$$\text{Approach 1: } \hat{\theta} = \frac{1}{n} \sum (X_i/3), \quad \sigma = \sqrt{\text{Var}(X_1/3)} = \sqrt{\frac{3\theta^2}{9}} = \frac{\theta}{\sqrt{3}}.$$

$$\text{Approach 2: } \sigma = \frac{1}{\sqrt{I(\theta)}} = \frac{1}{\sqrt{3/\theta^2}} = \frac{\theta}{\sqrt{3}}.$$

$\sigma = \theta/\sqrt{3}$. Both agree, confirming the CR bound is attained.

Worked Practice Problems

Practice 1: Weibull(2, θ) — Likelihood

Distribution. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x; \theta) = \frac{2x}{\theta^2} e^{-x^2/\theta^2}$, $x > 0$, $\theta > 0$.

Key structural observation. The exponent is $-x^2/\theta^2$ (not $-x/\theta$). This means: (1) the collapsed exponential in L will be $e^{-\sum X_i^2/\theta^2}$, and (2) the correct u-substitution is $u = x^2/\theta^2$.

Likelihood:

$$L(\theta) = \prod_{i=1}^n \frac{2X_i}{\theta^2} e^{-X_i^2/\theta^2} = \frac{2^n}{\theta^{2n}} \cdot \prod_{i=1}^n X_i \cdot \exp\left(-\frac{\sum X_i^2}{\theta^2}\right).$$

Practice 2: Rayleigh(θ) — MLE and Second Derivative

Distribution. $f(x; \theta) = \frac{2x}{\theta} e^{-x^2/\theta}$, $x > 0$. (Note: parameter θ appears as θ , not θ^2 , in the exponent — watch the powers carefully.)

Log-likelihood:

$$\ell(\theta) = n \log 2 - n \log \theta + \sum \log X_i - \frac{\sum X_i^2}{\theta}.$$

Score equation $\ell'(\theta) = 0$:

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum X_i^2}{\theta^2} = 0 \implies \hat{\theta} = \frac{\sum X_i^2}{n}.$$

Second derivative test:

$$\ell''(\theta) = \frac{n}{\theta^2} - \frac{2 \sum X_i^2}{\theta^3}.$$

From the score equation: $\sum X_i^2 = n\hat{\theta}$. Therefore:

$$\ell''(\hat{\theta}) = \frac{n}{\hat{\theta}^2} - \frac{2n\hat{\theta}}{\hat{\theta}^3} = \frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} = -\frac{n}{\hat{\theta}^2} < 0. \quad \checkmark$$

Practice 3: Gamma(3, θ) — Full Q1–Q8 Chain

Distribution. $f(x; \theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$, $x > 0$. (Gamma with shape $\alpha = 3$, scale θ .)

Quantity	Computation	Result
$L(\theta)$	$\prod f(X_i; \theta)$	$(2\theta^3)^{-n} \prod X_i^2 \cdot e^{-\sum X_i/\theta}$
$\hat{\theta}$	$\ell'(\theta) = 0$	$\bar{X}/3 = \sum X_i/(3n)$
$\ell''(\hat{\theta})$	$\text{sub } \sum X_i = 3n\hat{\theta}$	$-3n/\hat{\theta}^2 < 0 \checkmark$
$\mathbb{E}[X_1]$	$u = x/\theta$, row $n = 3$	3θ
$\text{Bias}(\hat{\theta}, \theta)$	$\mathbb{E}[\hat{\theta}] - \theta = \theta - \theta$	0 (unbiased)
$\mathbb{E}[X_1^2]$	$u = x/\theta$, row $n = 4$	$12\theta^2$
$\text{Var}(X_1)$	$12\theta^2 - (3\theta)^2$	$3\theta^2$
$\text{MSE}(\hat{\theta}, \theta)$	$\text{Var}(X_1)/(9n)$	$\theta^2/(3n)$
Consistency	Chebyshev: $\theta^2/(3n\varepsilon^2) \rightarrow 0$	$\hat{\theta} \xrightarrow{p} \theta$
$I(\theta)$	$\text{Var}(X_1)/\theta^4$	$3/\theta^2$
CRLB	$1/(nI(\theta))$	$\theta^2/(3n)$
Bound attained?	$\text{Var}(\hat{\theta}) = \text{CRLB}$	Yes — UMVUE
σ (Q8)	$1/\sqrt{I(\theta)}$	$\theta/\sqrt{3}$

Practice 4: Exponential(λ) — Fisher Information

Distribution. $f(x; \lambda) = \lambda e^{-\lambda x}$, $x > 0$.

Score function:

$$\log f = \log \lambda - \lambda X_1, \quad s(\lambda) = \frac{1}{\lambda} - X_1 = -(X_1 - \underbrace{\mathbb{E}[X_1]}_{=1/\lambda}).$$

Fisher information (using the variance shortcut):

$$I(\lambda) = \text{Var}(X_1) = \frac{1}{\lambda^2}.$$

CRLB and σ :

$$\text{CRLB} = \frac{1}{n \cdot 1/\lambda^2} = \frac{\lambda^2}{n}, \quad \sigma = \frac{1}{\sqrt{I(\lambda)}} = \lambda.$$

Practice 5: Uniform($0, \theta$) — Bound Not Attained

Distribution. $f(x; \theta) = 1/\theta$, $0 < x < \theta$. **Support depends on θ** — regularity conditions fail.

MLE. $L(\theta) = \theta^{-n} \mathbf{1}(X_{(n)} \leq \theta)$ is decreasing in θ for $\theta \geq X_{(n)}$, so $\hat{\theta} = X_{(n)}$.

Bias. The pdf of $X_{(n)}$ is $f_{(n)}(x) = nx^{n-1}/\theta^n$. Then:

$$\mathbb{E}[X_{(n)}] = \frac{n}{\theta^n} \int_0^\theta x^n dx = \frac{n\theta}{n+1}, \quad \text{Bias} = -\frac{\theta}{n+1}.$$

Bias-corrected estimator. $T^* = \frac{n+1}{n} X_{(n)}$ is unbiased. Its variance:

$$\text{Var}(T^*) = \frac{\theta^2}{n(n+2)}.$$

Formal CRLB. With $s(\theta) = -1/\theta$ (constant in X_1 !) and $I(\theta) = 1/\theta^2$:

$$\text{CRLB} = \frac{\theta^2}{n}.$$

Comparison.

$$\text{Var}(T^*) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n} = \text{CRLB}.$$

T^* **beats the formal CRLB** — possible only because the regularity conditions fail. The bound is not a valid lower bound for this model.

Consistency: Two Approaches Compared

Both approaches prove $\hat{\theta} \xrightarrow{p} \theta$ but differ in what they require.

	Law of Large Numbers	Chebyshev / MSE
Best when	$\hat{\theta} = (1/n) \sum g(X_i)$	MSE already computed (Q4 done)
Requires	$\mathbb{E}[g(X_1)] = \theta; \mathbb{E}[g(X_1)] < \infty$	$\text{MSE}(\hat{\theta}, \theta) \rightarrow 0$
Key step	LLN: $(1/n) \sum g(X_i) \xrightarrow{p} \mathbb{E}[g(X_1)] = \theta$	$P(\hat{\theta} - \theta \geq \varepsilon) \leq \text{MSE} / \varepsilon^2 \rightarrow 0$
Fails when	$\hat{\theta}$ not an average (e.g. $X_{(n)}$, ratio)	MSE doesn't vanish
Example	$\hat{\theta} = \bar{X}/3 = (1/n) \sum (X_i/3)$	$\hat{\theta} = \sqrt{\sum X_i^2 / n}$

Panaretos Remark 3.8: $\text{MSE}(\hat{\theta}, \theta) \rightarrow 0$ implies $\hat{\theta} \xrightarrow{p} \theta$. Therefore if you've computed MSE in Q4 and it goes to zero, consistency (Q5) is essentially already proven.

The Three-Way Identity

When the CR bound is attained, the following quantities are all equal:

$$\boxed{\text{Var}(g(X_1)) = \frac{1}{I(\theta)} = n \cdot \text{Var}(\hat{\theta}) = n \cdot \text{CRLB} = \sigma^2}$$

This identity simultaneously explains why:

1. The classical CLT and the MLE CLT give the *same* σ for Q8.
2. The UMVUE achieves the exact lower bound.
3. $\hat{\theta}$ is an *efficient* estimator.

When the bound is *not* attained (e.g. $\text{Uniform}(0, \theta)$), the identity breaks, the two CLT routes diverge, and the standard MLE CLT no longer applies.

Mock Exam: Weibull(2, θ) — Full Q1–Q8

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x; \theta) = \frac{2x}{\theta^2} e^{-x^2/\theta^2}$, $x > 0$, $\theta > 0$. Work all eight questions. The formula sheet in Section 1 is available.

Q6.1. Likelihood $L(\theta)$

Write and simplify $L(\theta) = \prod_{i=1}^n f(X_i; \theta)$.

Solution.

$$L(\theta) = \frac{2^n}{\theta^{2n}} \cdot \prod_{i=1}^n X_i \cdot \exp\left(-\frac{\sum X_i^2}{\theta^2}\right).$$

Group: $(2/\theta^2)^n$ from the constant; $\prod X_i$ from the data; $\exp(-\sum X_i^2/\theta^2)$ from collapsing the exponentials.

Q6.2. MLE $\hat{\theta}$ and second derivative test

Solution.

$$\begin{aligned}\ell(\theta) &= n \log 2 - 2n \log \theta + \sum \log X_i - \frac{\sum X_i^2}{\theta^2}. \\ \ell'(\theta) &= -\frac{2n}{\theta} + \frac{2 \sum X_i^2}{\theta^3} = 0 \implies \hat{\theta} = \sqrt{\frac{\sum X_i^2}{n}}. \\ \ell''(\theta) &= \frac{2n}{\theta^2} - \frac{6 \sum X_i^2}{\theta^4}.\end{aligned}$$

From score equation: $\sum X_i^2 = n\hat{\theta}^2$. Then: $\ell''(\hat{\theta}) = 2n/\hat{\theta}^2 - 6n/\hat{\theta}^2 = -4n/\hat{\theta}^2 < 0$. ✓

Q6.3. Bias via $\mathbb{E}[X_1^2]$

Solution. The natural unbiased estimator is $\tilde{\theta}^2 = \sum X_i^2/n$ for the target θ^2 . Compute:

$$\mathbb{E}[X_1^2] = \frac{2}{\theta^2} \int_0^\infty x^3 e^{-x^2/\theta^2} dx.$$

Substitute $u = x^2/\theta^2$, so $x^2 = \theta^2 u$ and $2x dx = \theta^2 du$, i.e. $x^3 dx = \theta^2 u \cdot \frac{\theta^2}{2} du = \frac{\theta^4}{2} u du$:

$$\mathbb{E}[X_1^2] = \frac{2}{\theta^2} \cdot \frac{\theta^4}{2} \int_0^\infty u e^{-u} du = \theta^2 \cdot 1 = \theta^2.$$

Therefore $\mathbb{E}[\tilde{\theta}^2] = \theta^2$, so $\text{Bias}(\tilde{\theta}^2, \theta^2) = 0$.

Q6.4. MSE of $\tilde{\theta}^2$

Solution.

$$\mathbb{E}[X_1^4] = \frac{2}{\theta^2} \int_0^\infty x^5 e^{-x^2/\theta^2} dx.$$

Same substitution: $x^5 dx = (x^2)^2 x dx = \theta^4 u^2 \cdot \frac{\theta^2}{2} du = \frac{\theta^6}{2} u^2 du$.

$$\mathbb{E}[X_1^4] = \frac{2}{\theta^2} \cdot \frac{\theta^6}{2} \cdot 2! = 2\theta^4.$$

$$\text{Var}(X_1^2) = 2\theta^4 - (\theta^2)^2 = \theta^4, \quad \text{Var}(\tilde{\theta}^2) = \frac{\theta^4}{n}, \quad \text{MSE}(\tilde{\theta}^2, \theta^2) = \boxed{\frac{\theta^4}{n}}.$$

Q6.5. Consistency of $\tilde{\theta}^2$

Solution.

LLN: $\tilde{\theta}^2 = (1/n) \sum X_i^2 \xrightarrow{p} \mathbb{E}[X_1^2] = \theta^2$. ✓

Chebyshev: $P(|\tilde{\theta}^2 - \theta^2| \geq \varepsilon) \leq \theta^4/(n\varepsilon^2) \rightarrow 0$. ✓

Q6.6. Fisher Information $I(\theta)$ **Solution.**

$$\log f = \log 2 + \log x - 2 \log \theta - \frac{x^2}{\theta^2}, \quad s(\theta) = -\frac{2}{\theta} + \frac{2X_1^2}{\theta^3} = \frac{2}{\theta^3}(X_1^2 - \theta^2).$$

Since $\mathbb{E}[X_1^2] = \theta^2$, the score is $(2/\theta^3)(X_1^2 - \mathbb{E}[X_1^2])$:

$$I(\theta) = \frac{4}{\theta^6} \text{Var}(X_1^2) = \frac{4}{\theta^6} \cdot \theta^4 = \frac{4}{\theta^2}.$$

Q6.7. Does $\tilde{\theta}^2$ attain the CR bound?**Solution.** Estimating $g(\theta) = \theta^2$, so $g'(\theta) = 2\theta$:

$$\text{CRLB} = \frac{[g'(\theta)]^2}{n \cdot I(\theta)} = \frac{4\theta^2}{n \cdot 4/\theta^2} = \frac{\theta^4}{n}.$$

Since $\text{Var}(\tilde{\theta}^2) = \theta^4/n = \text{CRLB}$, the bound **is attained**. $\tilde{\theta}^2$ is the UMVUE for θ^2 .

Q6.8. Asymptotic Normality — find σ **Solution.**

Approach 1 (Classical CLT): $\tilde{\theta}^2 = (1/n) \sum X_i^2$, so $Y_i = X_i^2$ are i.i.d. with $\text{Var}(Y_1) = \theta^4$.

$$\sigma = \sqrt{\text{Var}(X_1^2)} = \theta^2.$$

Approach 2 (MLE CLT): $\sigma = g'(\theta)/\sqrt{I(\theta)} = 2\theta/\sqrt{4/\theta^2} = 2\theta \cdot \theta/2 = \theta^2$.

Both agree: $\boxed{\sigma = \theta^2}$, consistent with the CR bound being attained.

Quick Reference Summary

Q1 — Likelihood: $L(\theta) = c(\theta)^n \cdot \prod g(X_i) \cdot \exp(-h(\sum X_i^k)/\theta^m)$

Q2 — MLE: $\ell'(\hat{\theta}) = 0$; substitute $\sum(\text{data}) = n \cdot g(\hat{\theta})$ into $\ell''(\hat{\theta})$ to show < 0 .

Q3 — Bias: $\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$; reduce to $\mathbb{E}[X_1]$ via linearity; u-sub $u = x/\theta$ or $u = x^2/\theta^2$; use $\int_0^\infty u^n e^{-u} du = n!$

Q4 — MSE: $\text{MSE} = \text{Bias}^2 + \text{Var}(\hat{\theta})$; $\text{Var}(\hat{\theta}) = \text{Var}(X_1)/(c^2 n)$; need $\mathbb{E}[X_1^2]$ (one row higher in table).

Q5 — Consistency: LLN if $\hat{\theta} = (1/n) \sum g(X_i)$ and $\mathbb{E}[g(X_1)] = \theta$; or Chebyshev if $\text{MSE} \rightarrow 0$.

Q6 — Fisher info: $I(\theta) = \text{Var}(s(\theta))$ where $s = \frac{\partial}{\partial \theta} \log f(X_1; \theta)$; shortcut: if $s = (X_1 - \mathbb{E}[X_1])/c(\theta)$ then $I(\theta) = \text{Var}(X_1)/c(\theta)^2$.

Q7 — CR bound: $\text{CRLB} = 1/(nI(\theta))$ for unbiased estimators; $[g'(\theta)]^2/(nI(\theta))$ for estimating $g(\theta)$. Attained iff $\text{Var}(\hat{\theta}) = \text{CRLB}$. If support depends on θ : regularity fails, bound may be invalid.

Q8 — Asymptotic normality: Classical CLT: $\sigma = \sqrt{\text{Var}(g(X_1))}$. MLE CLT: $\sigma = 1/\sqrt{I(\theta)}$. These agree iff bound attained.

- ☐ $L(\theta)$: grouped correctly? Exponent collapsed to $\exp(-\sum X_i^k/\theta^m)$?
- ☐ MLE: second derivative test done and $\ell''(\hat{\theta}) < 0$ shown explicitly?
- ☐ Bias: u-substitution tracks all four pieces (substitution, x^k , $e^{-(\cdot)}$, dx)?
- ☐ MSE: n appears in the denominator?
- ☐ Consistency: stated the conclusion " $\hat{\theta} \xrightarrow{p} \theta$ " explicitly?
- ☐ Fisher info: used $\mathbb{E}[s(\theta)] = 0$ so $I(\theta) = \text{Var}(s)$? No new integrals?
- ☐ CR bound: computed CRLB; stated whether attained; cited regularity if not?
- ☐ Q8: wrote both CLT approaches; confirmed they agree; stated σ clearly?